Recovering an effect nothing in the data can deconfound

Lifestyle and genetics drive both the exposure and the disease, and nobody measured them. No covariate adjustment can work. Is the effect gone, or is there another way in?

Abstract

Objective. Lifestyle and genetics drive both the exposure and the disease, and nobody measured them. No covariate adjustment can work. Is the effect gone, or is there another way in? **Methods.** First establish that the obvious route is genuinely closed. 5 step(s) are recorded in full below. **Results.** The naive estimate was 2.30 against a truth of 1.80; the front-door estimate was 1.81. Nothing was measured in between. The gap of +0.50 is the confounding, now quantified by a study that never observed the confounder. **Conclusions.** No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: Lifestyle and genetics drive both the exposure and the disease, and nobody measured them. No covariate adjustment can work. Is the effect gone, or is there another way in?

2. Methods

Question. Lifestyle and genetics drive both the exposure and the disease, and nobody measured them. No covariate adjustment can work. Is the effect gone, or is there another way in?

First establish that the obvious route is genuinely closed

Before reaching for anything clever it is worth confirming there is nothing to be had the ordinary way. On a graph with only the exposure, the outcome and an unmeasured common cause, identify() refuses — and the refusal is typed, with a reason, rather than being an empty result the caller has to interpret.

Considered instead. Assuming the problem is a missing covariate and going looking for proxies. Proxies for an unmeasured confounder do not close the back door unless they close it completely, and 'we adjusted for a rich set of covariates' is the sentence that most often stands in for a route nobody checked.

```
graph : X -> Y, X <-> Y
status : blocked
reason : no back-door, front-door, or instrumental route exists in the
graph
```

Add the one structural claim that changes everything

The route works by splitting the problem in two. Exposure to mediator is unconfounded, because the hidden common cause does not touch the mediator directly. Mediator to outcome is deconfoundable by conditioning on the exposure. Neither half needs the confounder, so their product does not either — this is the structure behind the smoking-tar-cancer argument. The claim is that the exposure reaches the outcome only through a measured intermediate — a biomarker, a deposited dose. That is not an extra assumption axiom bolts on; it IS the graph, and the graph is hashed into the verdict, so a result can never later be quoted against a different one. With the mediator present the front-door route opens and the effect becomes recoverable despite the confounder still being invisible.

Considered instead. An instrument. It would work too, but it requires finding something that moves the exposure and touches the outcome no other way — and in observational epidemiology that thing usually does not exist, while the mediator often has already been measured for other reasons.

```
graph : M -> Y, X -> M, X <-> Y
measured : ['M', 'X', 'Y']
status : identified
route : frontdoor
mediators : ['M']
graph hash : 816f14bebf6c0f74...
```

Estimate it both ways on the same 8,000 people

The naive regression is what a careful analyst would report from these columns if nobody had drawn the graph, and it is confidently wrong. The front-door estimate uses no additional data at all — same rows, same columns — only the structural claim.

Considered instead. Reporting the front-door number alone. The distance between the two is the size of the confounding, which is a quantity the study can now put a number on despite never having measured the confounder.

```
naive (confounded) 2.296 +/- 0.015 [ 2.266, 2.327] error +0.496
front-door 1.810 +/- 0.020 [ 1.770, 1.850] error +0.010
```

```
truth 1.800
```

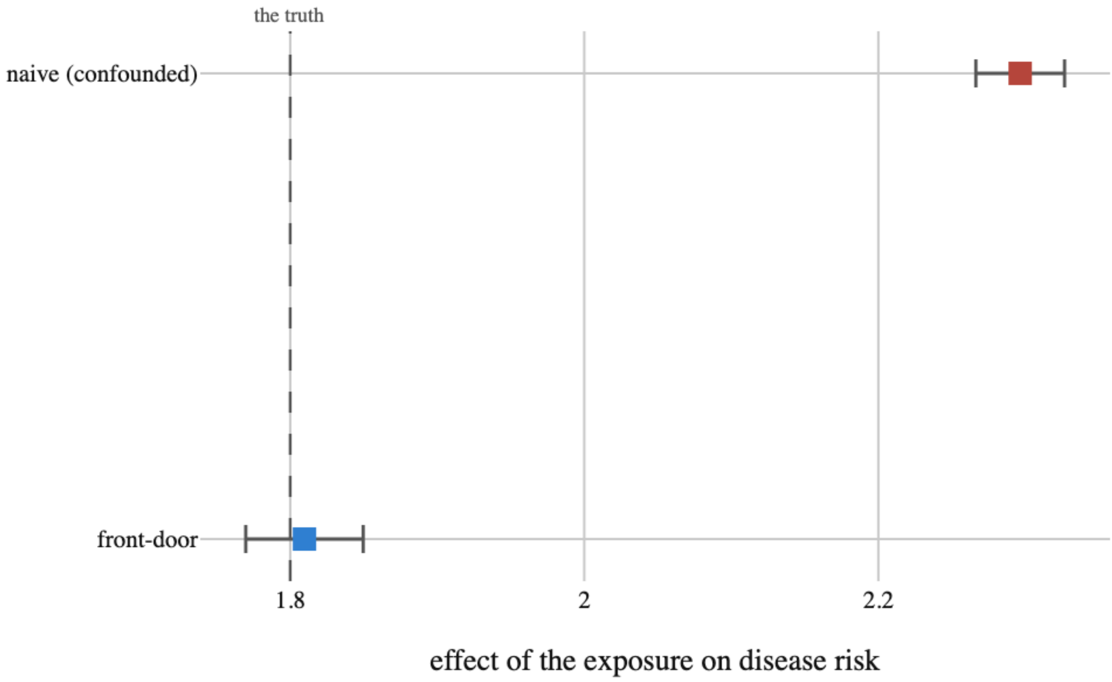

Figure 1. Same rows, same columns, one structural claim apart. No covariate adjustment could have produced the lower interval, because the confounder is not in the data. What produced it was a claim about how the world is wired

Break the claim on purpose and watch the route disappear

A load-bearing assumption should be shown bearing the load. Adding a direct exposure-to-outcome edge — a second pathway that skips the mediator — leaves a graph with no identifying route at all, and identify() says so rather than falling back on the front-door formula regardless.

Considered instead. Testing the assumption against the data. It cannot be tested against the data: both graphs imply the same joint distribution over the measured variables here. That is exactly why it has to be argued substantively and recorded explicitly.

```
graph : M -> Y, X -> M, X -> Y, X <-> Y
status : blocked
reason : no back-door, front-door, or instrumental route exists in the
graph
```

And measure what believing it wrongly would cost

Refusing on a graph is one thing; knowing the size of the mistake is another. Here the same analysis runs on a family of worlds that each have a small direct path the analyst does not know about, so the exclusion claim is false by a controlled amount. The front-door estimator keeps returning the mediated part and misses the direct part entirely.

Considered instead. Treating the assumption as binary — 'it holds' or 'the analysis is invalid'. Assumptions fail by degrees, and an estimator that degrades smoothly is usable under a claim that is nearly true, which is the only kind of claim epidemiology ever gets.

hidden direct path	true total effect	front-door says	error
0.00	1.800	1.810	+0.010

0.10	1.900	1.810	-0.090
0.20	2.000	1.810	-0.190
0.35	2.150	1.810	-0.340
0.50	2.300	1.810	-0.490
0.75	2.550	1.810	-0.740
1.00	2.800	1.810	-0.990

Table 1

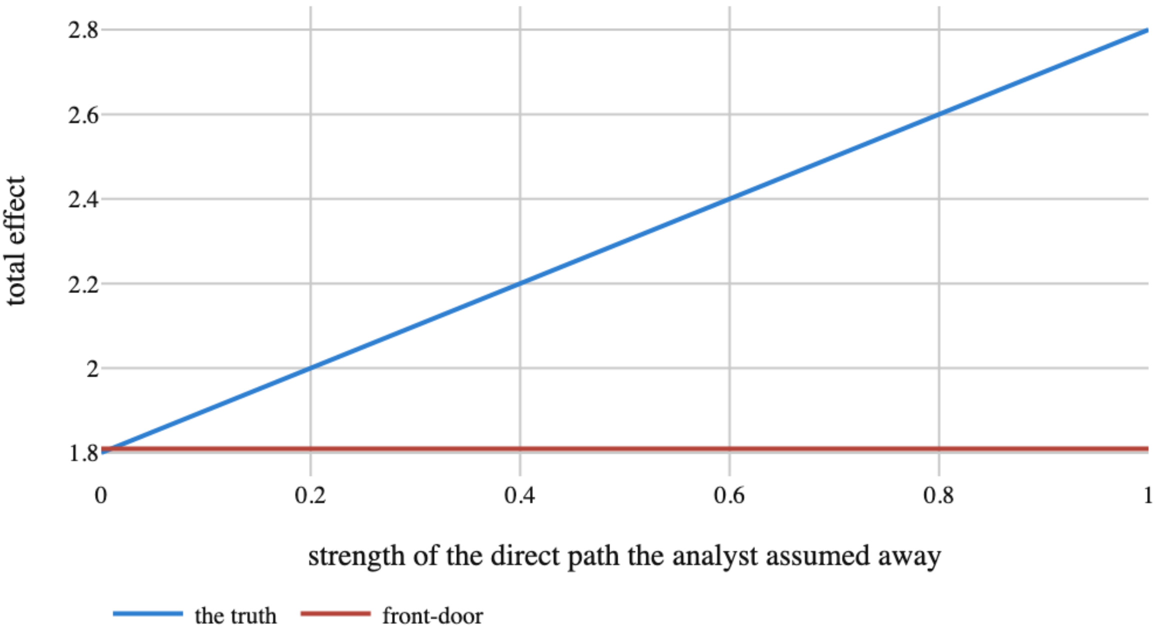

Figure 2. How wrong the answer gets as the exclusion claim fails. The front-door line is flat: it recovers the mediated part correctly and is blind to everything else, so the error is exactly the size of the path that was assumed away. That is a good failure mode — it is bounded by something a subject expert can reason about

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

The naive estimate was 2.30 against a truth of 1.80; the front-door estimate was 1.81. Nothing was measured in between. The gap of +0.50 is the confounding, now quantified by a study that never observed the confounder.

What made it recoverable was a structural claim, not a better estimator. That is the pattern worth taking away: when the answer is not identified from what you measured, the fix is a defensible claim about the world or a new measurement. A more elaborate fit is never the fix, because the problem is not in the fitting.

And the claim really is load-bearing. Add a direct edge and the graph offers no route at all; leave the edge in the world but out of the analysis and the error is the size of the edge — -0.99 when the direct path is as strong as 1.0. Both behaviours are what you want: refuse when the structure is wrong, degrade proportionally when it is only slightly wrong.

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

Lifestyle and genetics drive both the exposure and the disease, and nobody measured them. No covariate adjustment can work. Is the effect gone, or is there another way in?

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.